Lane E. Schultz, Ph.D.

Materials scientist and engineer

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Materials scientist combining uncertainty-aware machine learning, Bayesian optimization, and multiscale simulation to accelerate materials discovery and connect structure to properties.

Research Experience

Postdoctoral Appointee Dec 2024 – Present

Center for Integrated Nanotechnologies, Sandia National Laboratories | Albuquerque, NM

- **Metal–organic framework discovery:** Developed graph neural network surrogates with uncertainty estimates and multi-objective Bayesian optimization to screen frameworks for volatile organic compound separations.
- **Composite electrolyte screening:** Developed multi-objective Bayesian optimization for a pool of 174,758 LLZO–polymer microstructures, comparing ensemble linear regression and Gaussian-process surrogates across transport, mechanical, and dendrite-growth objectives.
- **Sputtering and deposition modeling:** Investigated surface-binding-energy distributions in crystalline and amorphous Mo–S systems and their effects on collision and transport simulations benchmarked against focused-ion-beam sputter-yield measurements.

Research Assistant Jun 2018 – Aug 2024

Computational Materials Group, University of Wisconsin–Madison | Madison, WI

- **Reliable materials machine learning:** Developed a kernel-density-estimation method to assess model applicability domains, connecting chemical dissimilarity with prediction errors and unreliable uncertainty estimates.
- **Metallic glass forming ability:** Developed high-throughput workflows for training machine learning interatomic potentials and calculating simulation-derived features to predict glass forming ability.
- **Research computing infrastructure:** Helped build and administer two high-performance computing clusters. Compiled research software and implemented Environment Modules for reproducible software environments.

Education

GPAs: Ph.D. and M.S. 3.70/4.00, and B.S. 3.99/4.00.

Ph.D., Materials Science and Engineering Aug 2024
University of Wisconsin–Madison, with a distributed minor in Machine Learning

M.S., Materials Science and Engineering Dec 2020
University of Wisconsin–Madison

B.S., Engineering Dec 2017
Fort Lewis College, with a concentration in Electro/Mechanical Engineering and a minor in Mathematics

Technical Expertise

Machine learning: PyTorch, scikit-learn, graph neural networks, Bayesian optimization, uncertainty quantification and applicability domains
Scientific computing: Python, C++, MATLAB, VASP, LAMMPS
Research infrastructure: Linux, Bash, Git, OpenHPC, Environment Modules, Docker, Apptainer, Vim, LaTeX
Engineering tools: AutoCAD, SolidWorks, Verilog, SPICE

Publications

Author name in bold. Journal articles use the final journal-issue year where available.

Journal Articles

1. **L. E. Schultz**, N. Gantzler, N. S. Bobbitt, D. F. Sava Gallis, and R. Dingreville (2026). Graph neural network-based multi-objective Bayesian optimization for enhanced screening of metal–organic frameworks with optimal separation performance. *Journal of Materials Chemistry A* 14(18), pp. 10836–10853. DOI: 10.1039/D5TA09133K
2. **L. E. Schultz**, Y. Wang, R. Jacobs, and D. Morgan (2025). A general approach for determining applicability domain of machine learning models. *npj Computational Materials* 11(1), 95. DOI: 10.1038/s41524-025-01573-x
3. **L. E. Schultz**, B. Afflerbach, P. M. Voyles, and D. Morgan (2025). Machine learning metallic glass critical cooling rates through elemental and molecular simulation based featurization. *Journal of Materiomics* 11(4), 100964. DOI: 10.1016/j.jmat.2024.100964
4. J. Meng, M. S. Sheikh, **L. E. Schultz**, W. O. Nachlas, J. Liu, M. P. Polak, R. Jacobs, and D. Morgan (2025). Ultrafast Oxygen Conduction in Sillén Oxychlorides. *Chemistry of Materials* 37(9), pp. 3147–3157. DOI: 10.1021/acs.chemmater.4c03163
5. V. Agrawal, S. Zhang, **L. E. Schultz**, and D. Morgan (2025). Accelerating ensemble uncertainty estimates in supervised materials property regression models. *Computational Materials Science* 246, 113494. DOI: 10.1016/j.commatsci.2024.113494
6. R. Jacobs, **L. E. Schultz**, A. Scourtas, K. J. Schmidt, O. Price-Skelly, W. Engler, I. Foster, B. Blaiszik, P. M. Voyles, and D. Morgan (2024). Machine learning materials properties with accurate predictions, uncertainty estimates, domain guidance, and persistent online accessibility. *Machine Learning: Science and Technology* 5(4), 045051. DOI: 10.1088/2632-2153/ad95db
7. K. J. Schmidt, A. Scourtas, L. Ward, S. Wangen, M. Schwarting, I. Darling, E. Truelove, A. Ambadkar, R. Bose, Z. Katok, J. Wei, X. Li, R. Jacobs, **L. Schultz**, D. Kim, M. Ferris, P. M. Voyles, D. Morgan, I. Foster, and B. Blaiszik (2024). Foundry-ML – Software and Services to Simplify Access to Machine Learning Datasets in Materials Science. *Journal of Open Source Software* 9(93), 5467. DOI: 10.21105/joss.05467
8. **L. E. Schultz**, B. Afflerbach, I. Szlufarska, and D. Morgan (2022). Molecular dynamic characteristic temperatures for predicting metallic glass forming ability. *Computational Materials Science* 201, 110877. DOI: 10.1016/j.commatsci.2021.110877
9. B. T. Afflerbach, C. Francis, **L. E. Schultz**, J. Spethson, V. Meschke, E. Strand, L. Ward, J. H. Perepezko, D. Thoma, P. M. Voyles, I. Szlufarska, and D. Morgan (2022). Machine Learning Prediction of the Critical Cooling Rate for Metallic Glasses from Expanded Datasets and Elemental Features. *Chemistry of Materials* 34(7), pp. 2945–2954. DOI: 10.1021/acs.chemmater.1c03542
10. **L. E. Schultz**, B. Afflerbach, C. Francis, P. M. Voyles, I. Szlufarska, and D. Morgan (2021). Exploration of characteristic temperature contributions to metallic glass forming ability. *Computational Materials Science* 196, 110494. DOI: 10.1016/j.commatsci.2021.110494
11. B. T. Afflerbach, **L. Schultz**, J. H. Perepezko, P. M. Voyles, I. Szlufarska, and D. Morgan (2021). Molecular simulation-derived features for machine learning predictions of metal glass forming ability. *Computational Materials Science* 199, 110728. DOI: 10.1016/j.commatsci.2021.110728
12. J. Xi, G. Bokas, **L. E. Schultz**, M. Gao, L. Zhao, Y. Shen, J. H. Perepezko, D. Morgan, and I. Szlufarska (2020). Microalloying effect in ternary Al-Sm-X (X=Ag, Au, Cu) metallic glasses studied by ab initio molecular dynamics. *Computational Materials Science* 185, 109958. DOI: 10.1016/j.commatsci.2020.109958
13. B. R. Halls, J. R. Gord, **L. E. Schultz**, W. C. Slowman, M. D. A. Lightfoot, S. Roy, and T. R. Meyer (2018). Quantitative 10–50 kHz X-ray radiography of liquid spray distributions using a rotating-anode tube source. *International Journal of Multiphase Flow* 109, pp. 123–130. DOI: 10.1016/j.ijmultiphaseflow.2018.07.014

Manuscripts Under Review and in Preparation

M1. **L. E. Schultz**, A. J. Mings, M. L. Meyerson, N. Jackson, D. P. Adams, and R. Dingreville (2026). Atomistically Resolved Surface Binding Energies for Mo–S Systems: Implications for Sputter Yield and Film Deposition Modeling. Under review

M2. **L. E. Schultz**, S. Q. Monismith, N. M. Francis, J. D. McBrayer, and R. Dingreville (2026). Accelerating LLZO–Polymer Composite Electrolyte Microstructure Screening Through Multi-Objective Bayesian Optimization. In preparation

Preprints

P1. R. Jacobs, **L. E. Schultz**, and D. Morgan (2026). Kolmogorov-Arnold Networks Applied to Materials Property Prediction. Preprint. DOI: 10.48550/arXiv.2601.20097. arXiv: 2601.20097

P2. R. Jacobs, M. P. Polak, **L. E. Schultz**, H. Mahdavi, V. Honavar, and D. Morgan (2024). Regression with Large Language Models for Materials and Molecular Property Prediction. Preprint; revised April 2026. DOI: 10.48550/arXiv.2409.06080. arXiv: 2409.06080

Conference Proceedings

C1. **L. E. Schultz**, T. J. Cogger, R. Good, J. Schneider, R. Rothschild, and W. Nollet (2018). Design of Torsional Test Stand for Micro-Newton Force Detection. In: *2018 Aerodynamic Measurement Technology and Ground Testing Conference*. DOI: 10.2514/6.2018-3737

C2. J. Schneider, **L. E. Schultz**, S. Mancha, E. Hicks, and R. N. Smith (2016). Development of a portable water quality sensor for river monitoring from small rafts. In: *OCEANS 2016 MTS/IEEE Monterey*, pp. 1–10. DOI: 10.1109/OCEANS.2016.7761392

Presentations and Posters

1. **L. E. Schultz**, N. Gantzler, N. S. Bobbitt, D. F. Sava Gallis, and R. Dingreville (2026). *GNNs for Selection of Optimal Metal Organic-Frameworks*. Electronic Materials Conference. Conference presentation. Ann Arbor, MI
2. **L. E. Schultz**, N. Gantzler, N. S. Bobbitt, D. F. Sava Gallis, and R. Dingreville (2026). *GNNs for Selection of Optimal Metal Organic-Frameworks*. TMS 2026, The Minerals, Metals & Materials Society. Conference presentation. San Diego, CA
3. **L. E. Schultz**, N. Gantzler, N. S. Bobbitt, D. F. Sava Gallis, and R. Dingreville (2025). *Graph Neural Network-Driven Multi-Objective Bayesian Optimization for Discovering Metal-Organic Frameworks with Optimal Separation Performance*. Chesapeake Large Scale Analytics Conference. Poster. Annapolis, MD
4. **L. E. Schultz**, B. T. Afflerbach, and D. Morgan (2021). *Molecular Dynamic Characteristic Temperatures for Predicting Metallic Glass Forming Ability*. Materials Science & Technology. Conference presentation. Columbus, OH
5. **L. E. Schultz**, B. T. Afflerbach, and D. Morgan (2020). *Molecular Dynamics Features for Predicting Metallic Glass Critical Casting Thickness*. Virtual Materials Research Society Spring/Fall Meeting & Exhibit. Conference presentation

Earlier Research and Engineering Experience

Summer Undergraduate Research Experience May – Aug 2017

Computational Nuclear Engineering Research Group, UW–Madison | Madison, WI

Automated visualization of Direct Accelerated Geometry Monte Carlo (DAGMC) geometries in VisIt using Python, following PEP 8 conventions for collaborative development.

Capstone Design Project Sep 2016 – Apr 2017

Fort Lewis College | Durango, CO

Co-designed and built a micro-newton-force propulsion test stand. Implemented an electrostatic displacement mechanism and modeled the system response in MATLAB.

Summer Undergraduate Research Fellowship May – Aug 2016

Advanced Diagnostics and Propulsion Research Laboratory, Purdue University | West Lafayette, IN

Operated pressure vessels, X-ray tube sources, and high-speed cameras to study two-dimensional sprays and constructed a protective lead enclosure for the X-ray sources.

Undergraduate Design Project Dec 2015 – Apr 2016

Fort Lewis College | Durango, CO

Developed a waterproof, interchangeable sensor package for river monitoring from rafts and kayaks, measuring water temperature, oxidation–reduction potential, pH, time, and GPS position.

Professional Service and Memberships

Member, Computational Materials Science and Engineering Committee Current
The Minerals, Metals & Materials Society (TMS)

Workshop Volunteer Mar 15, 2026
Software Development in Materials Science and Engineering, TMS 2026.
Assisted participants during the in-person workshop.

Honor societies: Sigma Pi Sigma (Physics Honor Society) and Order of the Engineer

Professional Development

Artificial Intelligence in Materials Science and Engineering Oct 20–22, 2026
Workshop, The Minerals, Metals & Materials Society (TMS)

Building AI Agents with Multimodal Models Sep 10, 2025
Workshop, NVIDIA

Teaching and Mentoring

- Mentored undergraduate and graduate researchers and assisted research-group peers with software installation and cluster use.
- Supported graduate molecular-dynamics laboratory instruction and graded Thermodynamics of Solids.
- Served as an undergraduate teaching assistant for Thermal and Fluid Systems Laboratory, and for Engineering Fundamentals II (MATLAB).
- Worked as a public-school substitute teacher at elementary, middle, and high-school levels.

Awards and Honors

University of Wisconsin–Madison: PPG Fellowship and Ying Yu Chuang Graduate Support Award
Fort Lewis College: Deans' Council Freshman 4.0 Award and Certificate, and Freshman Chemistry Recognition Award
Earlier recognition: Renaissance Plaque from Manzano High School, and Chickasaw Honor Club Outstanding Academic Achievement Award and Patrick S. Gilmore Band Award from Menard High School

Additional Information

Languages: English and Spanish (fluent written and verbal)
Citizenship: United States, Colombia, and the Chickasaw Nation